

Lectures

Stat 151A: Linear Models

Lectures marked with a * are still in progress.

These lecture notes are meant to supplement the reading indicated in the course schedule.

Introduction

- [Course philosophy](#)
- [Sample means](#)
- [Linear algebra review](#)

Unit 1: Multilinear regression

- [Simple linear regression](#)
- [Multilinear regression](#)
- [Paths to least squares](#)
- Transforming the regressors
 - [One-hot encodings](#)
 - [Interacting continuous regressors](#)
 - [Supplement: Linear transformations](#)
- [Transforming the response variable](#)

Unit 2: Inference for linear regression

- [Confidence intervals \(review\)](#)
- [Modeling of OLS](#)
- [OLS under normality](#)

- Supplement: The multivariate normal
 - Supplement: The distribution of the residual variance estimator
 - Variable selection and F-tests
- OLS under non-normality
 - Supplement: Asymptotics under misspecification
- Statistical versus practical significance

Unit 3: Machine learning and regularization

- OLS for prediction with expressive basis functions
- Bayesian methods and L2 regularization

Unit 4: How regression can go wrong

- The FWL theorem
- Regression to the mean
- Omitted variables
- Influence and outliers